MA385: Assignment (Due **5pm, Friday 23 October**)

This assignment contributes 10% to your final grade for MA385. Your solutions must be clearly written (ideally, by hand), and neatly presented: up to 20% of the marks will be given for clarity and correctness of exposition.

For numerical calculations (such as in Q5(b)), you may make use a calculator or computer; details of the calculations are not required.

Write your name and ID number on the first page. Number every page. Upload a scanned PDF version to canvas at https://universityofgalway.instructure.com/courses/58642/assignments/165065.

If you really want, you can submit a hard copy at the lecture on the 22nd of October. If doing so, staple the pages together, and write your name and ID number at the top of each *page.*

The assignment is collaborative*: you are allowed work with classmates. However, each of you needs to submit your own hand-written version of your solutions, and you should include a clear statement on who you collaborated with, and on what aspects.*

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Q1. (Based on Q3 from Tutorial Sheet 1) Show that the Secant iteration can be written as

$$x_{k+1} = \frac{x_k f(x_{k-1}) - x_{k-1} f(x_k)}{\delta_k},$$

for $k = 1, 2, \ldots$, and give a formula for δ_k.

Q2. In Q5 of Tutorial Sheet 2, you deduced that, if the Secant Method computes the iterates $\{x_0, x_1, x_2, \ldots\}$ then

$$\tau - x_{k+1} = -(\tau - x_k)(\tau - x_{k-1}) \frac{f[x_{k-1}, x_k, \tau]}{f[x_{k-1}, x_k]}, \quad (1)$$

where

$$f[a, b] := \frac{f(b) - f(a)}{b - a}$$

and

$$f[a, b, c] := \frac{f[b, c] - f[a, b]}{c - a}.$$

(a) Use Equation (1) to give a justification (does not have to be rigorous) that there is a constant K such that, as $k \to \infty$, we get

$$|\tau - x_{k+1}| \approx K|\tau - x_k||\tau - x_{k-1}|.$$

(b) Use this to deduce that the Secant Method converges with at least order $q = (1 + \sqrt{5})/2$.

Q3. Suppose that we have a fixed point iteration (FPI) method $x_{k+1} = g(x_k)$ which we know to be converges to fixed point of g, denoted τ. Show that the method converges with at least order p if

$$g'(\tau) = g''(\tau) = \cdots = g^{(p-1)} = 0.$$

Q4. Suppose we want to estimate $\sqrt{3}$, via FPI using a method of the form

$$x_{k+1} = a x_k + b/x_k, \qquad \text{for } k = 0, 1, 2, \ldots. \quad (2)$$

Show that one needs to choose $b = 3 - 3a$.

Q5. Take $a = 3/4$ in Equation (2).

(a) Show that the resulting g is a contraction on $[1, 3]$.

(b) Take $x_0 = 1.5$, and compute the corresponding values of x_1, x_2 and x_3. Compute the errors $\mathcal{E}_k = |\tau - x_k|$ for $k = 0, 1, 2, 3$.

(c) Show that $\mathcal{E}_{k+1}/\mathcal{E}_k$ is roughly constant. What can we infer from that?

Q6. Determine the values of a and b in Eq (2) that would correspond to Newton's method applied to solving $x^2 - 3 = 0$. Show that this method converges with at least order 2.